LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Optical holography basic principle, procedure for setting up construction and re-construction, laser, interferogram advantages disadvantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Optical holography basic principle, procedure for setting up construction and re-construction, laser, interferogram advantages disadvantages and application
5. Specific Objectives	16.1.5.1 Trainee will be able to explain the optical holography and acoustic emission inspection

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Laser alignment and Interferogram formation; Safety precautions regarding laser eye protection and beam paths; Instructor supervision of trainee setup procedures
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the fundamental difference between a photograph and a hologram?
(b) Define the term 'interference' in the context of wave optics.
(c) Why is a coherent light source required for holography?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of optical holography • Role of reference and object beams • Formation of interference pattern • Recording medium requirements	L & Ds	PPT, IB	15 min
(b)	• Procedure for setting up construction • Laser alignment and beam splitting • Interferogram advantages and disadvantages • Re-construction of the hologram	L & Ds	PPT, PDF	20 min
(c)	• Applications of holography in inspection • Comparison with acoustic emission • Troubleshooting common setup errors • Safety protocols during operation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the re-construction process reveals the original wavefront.
(b) List two specific advantages of using interferograms for defect detection.
(c) Describe one critical safety precaution when handling the laser source.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of optical holography discussed in this lesson can be applied to the next lesson plan (LP 02) regarding advanced non-destructive testing methods.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Availability of specific holographic recording plates for practical demonstration.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min